

SECOND PUBLIC EXAMINATION

Honour School of Physics Part B: 3 and 4 Year Courses

Honour School of Physics and Philosophy Part B

B6. CONDENSED-MATTER PHYSICS

TRINITY TERM 2023

Wednesday, 7th June, 2.30 pm – 4.30 pm

*Answer **two** questions.*

*Start the answer to each question in a **fresh book**.*

The use of approved calculators is permitted.

A list of physical constants and conversion factors accompanies this paper.

The numbers in the margin indicate the weight that the Examiners expect to assign to each part of the question.

Do NOT turn over until told that you may do so.

1. The compound NaTaO_3 crystallises in a tetragonal structure with $a = 0.555$ nm and $c = 0.393$ nm. There are two distinct oxygen sites in the unit cell, labelled O1 and O2, and the ions have the following fractional coordinates:

Na: $0, \frac{1}{2}, \frac{1}{2}$ and $\frac{1}{2}, 0, \frac{1}{2}$

Ta: $0, 0, 0$ and $\frac{1}{2}, \frac{1}{2}, 0$

O1: $0, 0, \frac{1}{2}$ and $\frac{1}{2}, \frac{1}{2}, \frac{1}{2}$

O2: $\frac{1}{4} + \delta, -\frac{1}{4} + \delta, 0$; $-\frac{1}{4} - \delta, \frac{1}{4} - \delta, 0$; $\frac{1}{4} - \delta, \frac{1}{4} + \delta, 0$; and $-\frac{1}{4} + \delta, -\frac{1}{4} - \delta, 0$

where $\delta \ll 1$.

(a) Show that Na, Ta and O1 do not contribute to the intensity of a reflection with Miller indices hkl if $h + k \neq 2n$ (n is any integer), and determine the contribution of these ions to the structure factor in the case $h + k = 2n$. [5]

(b) Expand the structure factor contribution of O2 to the first order in δ and write expressions in this limit for the following cases: (i) h and k both odd (ii) h and k both even (iii) h or k even, the other odd. [8]

(c) A medium-resolution neutron powder diffraction experiment is performed on NaTaO_3 , with a wavelength $\lambda = 0.159$ nm and an angular resolution of 0.05° in 2θ . The neutron scattering (Fermi) lengths for Na, Ta and O are $b_{\text{Na}} = 3.63$ fm, $b_{\text{Ta}} = 6.91$ fm and $b_{\text{O}} = 5.803$ fm. For the Bragg peaks (001), (110) and (210) calculate (i) the square of the structure factor, (ii) the multiplicity, and (iii) the scattering angle 2θ . [Hint: NaTaO_3 is *not* cubic, so you must use the general formula for orthogonal axes $d = ((h/a)^2 + (k/b)^2 + (l/c)^2)^{-1/2}$ and the multiplicities as appropriate for the tetragonal symmetry].

Also verify that (001) and (110) are too close to be resolved in this experiment and will appear as a single peak. [8]

(d) The following intensities in arbitrary units are measured, already corrected for any geometrical factors: $I_{110} + I_{001} = 1843$, $I_{210} = 345$. Use this information to determine the value of δ . [4]

2. Within the framework of the Molecular Orbital (MO) theory, a trial wavefunction for a system of two identical atoms can be written as $|\psi\rangle = c_1|1\rangle + c_2|2\rangle$.

$|1\rangle$ and $|2\rangle$ are normalised atomic orbitals with the same energy ϵ_0 , centred at atoms 1 and 2, so that in the absence of the other atom, $(\hat{K} + \hat{V}_1)|1\rangle = \epsilon_0|1\rangle$ and $(\hat{K} + \hat{V}_2)|2\rangle = \epsilon_0|2\rangle$, while c_1 and c_2 are (generally complex) coefficients. Here, $\hat{K}$, $\hat{V}_1$ and $\hat{V}_2$ have the usual meaning. Approximate eigenstates for the 2-atom Hamiltonian $\hat{H} = \hat{K} + \hat{V}_1 + \hat{V}_2$ are sought by finding the extrema (minima, maxima and saddle points) of the expression $\lambda = \langle\psi|\hat{H}|\psi\rangle / \langle\psi|\psi\rangle$.

(a) *Without* making the approximation $\langle i|j\rangle = \delta_{ij}$, show that extrema are solutions of the 2×2 matrix equation:

$$\left[\begin{pmatrix} \epsilon_0 + V_{c2} & \sigma \epsilon_0 - t \\ \sigma^* \epsilon_0 - t^* & \epsilon_0 + V_{c1} \end{pmatrix} - \lambda \begin{pmatrix} 1 & \sigma \\ \sigma^* & 1 \end{pmatrix} \right] \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = 0,$$

where $V_{c1} = \langle 2|V_1|2\rangle$, $V_{c2} = \langle 1|V_2|1\rangle$, $t = -\langle 1|V_2|2\rangle$ and $\sigma = \langle 1|2\rangle$. [Hint: you can use without proof the ‘formal’ differentiation with respect to c_1^* and c_2^* or differentiate with respect to the real and imaginary parts of c_1 and c_2 .] [7]

(b) Assume that atoms are at positions $(d/2, 0, 0)$ and $(-d/2, 0, 0)$ and that the system is unchanged under $x \rightarrow -x$. (i) Show that in this case $V_{c1} = V_{c2}$, and t and σ are both *real*. (ii) By calculating $\begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$, (which for a matrix of this form does not depend on λ) or by any other method, determine the two approximate eigenstates of $\hat{H}$. [5]

(c) The ground-state wavefunction for the hydrogen atom is $\phi(r) = 1/\sqrt{\pi}e^{-r}$, where r is expressed in units of the Bohr radius a_0 . With these wavefunctions, the three integrals have exact expressions:

$$\begin{aligned} t &= 2e^{-d}(1+d) \\ \sigma &= e^{-d}(1+d+d^2/3) \\ V_c &= 2(e^{-2d}(1+d)-1)/d \end{aligned}$$

where $V_c \equiv V_{c1} = V_{c2}$, the energies are expressed in units of the Rydberg energy E_R (the hydrogen ground state energy is $-1 E_R$) and d is in units of a_0 . The experimentally observed distance between the centres of the atoms in the hydrogen molecule gives $d_{\text{exp}} \approx 1.4$, while its dissociation energy (i.e. the energy required to separate the two atoms from d_{exp} to infinity) is $E_d \approx 0.33 E_R$. (i) Explain qualitatively why t and σ should decrease exponentially at large values of d while V_c should behave as $1/d$ in the same limit. (ii) Calculate the ground-state energy at d_{exp} with two electrons in the lowest eigenstate, with and without the orbital overlap σ and comment on whether σ can be neglected. (iii) Show that the ground-state energy at $d = d_{\text{exp}}$ is much larger (more negative) than the experimental value, and (iv) that its value at $d = 0$ is even larger (in fact this eigenvalue has a negative minimum at $d = 0$, though you do not have to show this). [9]

(d) Discuss the discrepancies between the experimental values stated above and the analytical MO calculation. Which term(s) have been omitted, and why would you expect them to modify the results in line with the experimental findings? [4]

3. The electronic structure of a 2-dimensional material in the vicinity of the Fermi energy has the following dispersion:

$$E = E_0 \pm c\hbar k$$

where $c = 1.0 \times 10^6 \text{ m s}^{-1}$, the $+$ and $-$ signs correspond to the energies of the conduction and valence bands, respectively, and $k = |\mathbf{k}|$.

(a) (i) Sketch energy surfaces as a function of k_x and k_y for the case $E_F = E_0$, labelling the valence and conduction bands, and (ii) discuss similarities and differences between this material and a direct-gap semiconductor with $E_g \rightarrow 0$. [4]

(b) Show that the general expression for the area density of electrons in the conduction band as a function of temperature is:

$$n_{\text{el}} = C_1 \beta^p F_1(\alpha)$$

and determine C_1 and p . Here, $\beta = (k_B T)^{-1}$, $\alpha = \tilde{\mu} \beta$, $\tilde{\mu} = \mu - E_0$, μ is the chemical potential, and k_B is the Boltzmann constant. $F_1(x)$ is the first Fermi integral function, where in general:

$$F_j(x) = \frac{1}{j!} \int_0^\infty dt \frac{t^j}{e^{t-x} + 1}$$

and $dF_j(x)/dx = F_{j-1}(x)$. [5]

(c) Show that the electronic specific heat due to electrons in the conduction band is

$$c_v^{\text{el}} = C_2 k_B^3 T^q [a F_2(\alpha) + b \alpha F_1(\alpha)]$$

and determine C_2 , q , a , and b . [7]

(d) (i) Show that when $\tilde{\mu} = 0$ the number of electrons in the conduction band is the same as the number of holes in the valence band and that these electrons and holes make equal contributions to the specific heat. (ii) Using the approximation

$$\alpha \rightarrow 0 : F_2(\alpha) \approx \frac{3\zeta}{4} + \frac{\pi^2 \alpha}{12} + \frac{1}{2} \alpha^2 \log 2$$

with $\zeta = 1.2021$, determine the expression for the *total* electronic specific heat (electrons + holes) as a function of temperature for the case $\tilde{\mu} = 0$, and calculate its value at 100 K. [5]

(e) Using the approximation $\alpha \rightarrow \infty : F_2(\alpha) \approx \frac{\alpha^3}{6} + \frac{\pi^2 \alpha}{6}$,

(i) determine the temperature dependence of n_{el} in the limit $\tilde{\mu} \gg k_B T$.

(ii) Show that, for $T \rightarrow 0$,

$$\mu \rightarrow E_0 + C_3 (n_{\text{el}})^r$$

and determine C_3 and r . [4]

4. The Hamiltonian of an ion with angular momentum quantum number j , single-ion easy-axis anisotropy along x and magnetic field B orthogonal to the anisotropy axis may be written as

$$\hat{H} = -D\hat{J}_x^2 - g\mu_B\hat{J}_zB$$

where $D > 0$ is the anisotropy constant, g is the gyromagnetic ratio, μ_B is the Bohr magneton, and $\hat{J}_x^2$ can be expressed as $\hat{J}_x^2 = \frac{1}{4}(\hat{J}_+^2 + \hat{J}_-^2) + \frac{1}{2}(\hat{J}^2 - \hat{J}_z^2)$.

(a) For the case $j = 1$, recalling that $\hat{J}_\pm|j, m\rangle = \sqrt{(j \mp m)(j \pm m + 1)}|j, m \pm 1\rangle$, show that the energy levels of $\hat{H}$ can be obtained by diagonalising the matrix:

$$\mathbf{H} = -D \begin{pmatrix} a & 0 & a \\ 0 & 2a & 0 \\ a & 0 & a \end{pmatrix} - g\mu_B B \begin{pmatrix} b & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -b \end{pmatrix}$$

where a and b are numerical coefficients to be determined. [7]

(b) Having determined the eigenvalues of the matrix $\mathbf{H}$, expand the energy levels of $\hat{H}$ to the lowest order in B and sketch their behaviour as a function of B . [4]

(c) Using the approximate energy levels from part (b), (i) write the partition function of an ensemble of N non-interacting ions with Hamiltonian $\hat{H}$ and (ii) obtain its magnetic moment $m = -dF/dB$ in the low-field limit. (iii) Show that the low-field susceptibility $\chi = \mu_0 dm/dB$ for such an ensemble is

$$\chi_\perp = \chi_C f_\perp(D\beta)$$

where $\chi_C = 2N\mu_0(\mu_B g)^2\beta/3$ is the Curie susceptibility for $j = 1$, $\beta = 1/k_B T$, k_B is the Boltzmann constant, μ_0 is the vacuum permeability, and $f_\perp(x)$ is a function to be determined. [6]

(d) The corresponding expression for the $j = 1$ susceptibility with magnetic field *parallel* to the anisotropy axis is $\chi_\parallel = \chi_C f_\parallel(D\beta)$ where $f_\parallel(x) = 3e^x/(1 + 2e^x)$. Show that (i) $\chi_\parallel$, $\chi_\perp$, and χ_C have the same asymptotic behaviour for $T \rightarrow \infty$; (ii) $\chi_\parallel > \chi_C$ at all temperatures; and (iii) $\chi_\perp$ is finite at $T = 0$. [4]

(e) The figure below displays the molar susceptibilities $\chi_\parallel$, $\chi_\perp$, and χ_C for an ion with $g = 2$. (i) Provide a physical justification for this behaviour and (ii) use these data to determine the value of D . [4]

